Lecture 1 (September 2nd)

Remark. The main subject of this course is Galois theory. The motivation of this theory is to find a general solution of polynomial equations. For example, the general solution of the quadratic equation

$$ax^2 + bx + c \;=\; 0$$

where $a \neq 0$, b and c are real numbers is given by

$$x \;=\; \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

To be more precise, we want to express a general solution of the equation

$$a_n x^n + a_{n-1} x^{n-1} + \ldots + a_0 = 0$$

where $a_i \in \boldsymbol{Q}$ and $a_n \neq 0$ using 4 basic operations (namely, $\pm$, $\times$, and $\div$), and n-th roots. Today, we know that, through Galois theory, this is not possible for $n \geq 5$.

In essence, Galois theory connects the theory of fields and the theory of groups. Consider the following example. The polynomial

$$x^4 + x^3 + x^2 + x + 1 = 0$$

We know how to solve this,

$$(x-1)(x^4 + x^3 + x^2 + x + 1) = x^5 - 1 = 0$$

and thus we choose a primitive 5th root of unity (say $\xi = \exp(2\pi i/5)$). Then the solutions are

$$(\xi^1, \xi^2, \xi^3, \xi^4)$$

We see that this group is homomorphic to $\boldsymbol{Z}/5\boldsymbol{Z}$, and that there's a certain relationship between the roots of this polynomial and a certain group.

Chapter 13. Field Extensions

Definition. (13.1) Let K be a field. A field extension of K is another field F endowed with an injective ring homomorphism $\phi : K \to F$. You are essentially identifying the field with a subfield of a larger ring F.

Remark. From now on, we use the following conventions:

(i) Every ring is a commutative ring with 1

(ii) A ring homomorphism sends 1 to 1

Remark. There is at most one ring homomorphism between two fields. Notice that $\ker \phi$ is an ideal (additive subgroup with the absorption property) of K, so that $\ker \phi$ is either 0 or K. Therefore, ϕ must be injective.

Lecture 2 (September 4th)

Definition. (13.1) Let K be a field. A field extension of K is a pair $(F, i : K \to F)$ where F is a field and i is an injective (ring) homomorphism. We draw field extension tower diagrams like the following:

$$\begin{array}{c} F \\ | \\ K \end{array}$$

Example. (13.2 $*$) (Algebraic extension) Let $f(t) \in \boldsymbol{Q}[t]$ be a nonzero irreducible polynomial. Then $\boldsymbol{Q}[t] \,/\, (f(t))$ is a field, and the composite map

$$\boldsymbol{Q} \to \boldsymbol{Q}[t] \to \boldsymbol{Q}[t] \,/\, (f(t))$$

is a field extension. This is the general way we would be obtaining field extensions.

$$\begin{array}{c} \boldsymbol{Q}[t] \,/\, (f(t)) \\ | \\ \boldsymbol{Q} \end{array}$$

Proof. The set $\boldsymbol{Q}[t] \,/\, (f(t))$ is a maximal ideal and since $f(t)$ is a nonzero irreducible element of $\boldsymbol{Q}[t]$ which is principle ideal domain. Therefore, the quotient $\boldsymbol{Q}[t] \,/\, (f(t))$ is a field. $\square$

Example. (13.3) (Transcendental extension) Let $\boldsymbol{Q}(t)$ denote the field of fractions of $\boldsymbol{Q}[t]$. Then the composition map $\boldsymbol{Q} \to \boldsymbol{Q}[t] \to \boldsymbol{Q}(t)$ determines a field extension.

Definition. (13.6) The characteristic of a field K ($\mathrm{char}(K)$) is the smallest nonnegative generator of the kernel of the unique ring homomorphism $\phi : \boldsymbol{Z} \to K$.

Lemma. (13.7) Let $F \,/\, K$ be a field extension. Then

$$\mathrm{char}(K) = \mathrm{char}(F)$$

Lemma. (13.8) Let F be a field. Then F contains a unqiue copy of $\boldsymbol{Q}$ or $\boldsymbol{Z} \,/\, p\boldsymbol{Z}$, where p is prime.

13.2 Finite Extensions and the Degree of an Extension

Definition. (13.10) The degree of a field extension F/K, denoted $[F : K]$, is the dimension of F as a vector space over K.

$$[F : K] = \dim_K(F)$$

Remark. If $i : K \to F$ is a field extension, then we can view F as a K-vector space through the map i.

Definition. (13.11) A field extension is finite if $[F : K] < \infty$.

Definition. (algebraic and transcendental elements) Let F/K be a field extension, and let $\alpha \in F$. We will say that α is algebraic over K if there exists a nonzero polynomial $f(t) \in K[t]$ such that $f(\alpha) = 0$. The element $\alpha \in F$ is transcendental if not algebraic.

Example. Consider

$$\begin{tikzcd} \boldsymbol{C} \arrow[d, dash] \\ \boldsymbol{R} \end{tikzcd}$$

(i) Is $i \in \boldsymbol{C}$ algebraic over $\boldsymbol{R}$? Yes! i is a zero of $t^2 + 1 \in \boldsymbol{R}[t]$.

(ii) $\sqrt{2} \in \boldsymbol{R}$ is algebraic over $\boldsymbol{Q}$

(iii) Also, $\pi, e \in \boldsymbol{C}$ is transcendental over $\boldsymbol{Q}$.

(iv) Every element α of K is algebraic over K as we have the polynomial

$$t - \alpha \in K[t]$$

Definition. (13.20) Let F/K be a field extension. If $\alpha \in F$ is algebraic, the minimal polynomial of α over K is the unique monic irreducible polynomial $p(t) \in K[t]$ such that $f(\alpha) = 0$.

Proof. We show that the minimal polynomial exists and is unique. We will achieve this by showing that it is exactly the generator of the kernel of the evaluation map. First consider the evaluation homomorphism

$$\mathrm{ev}_\alpha : K[t] \to F$$

where

$$f(t) \mapsto f(\alpha) \qquad \& \qquad \sum_{i=0}^{n} a_i t^i \mapsto \sum_{i=0}^{n} a_i \alpha^i$$

Observe that it is indeed a homomorphism as it preserves addition, multiplication, and the identity element. Then, notice the following:

(I) The kernal of the map is automatically an ideal, and is of the form $(p(t))$ as $K[t]$ is a principle ideal domain.

(II) By the 1st isomorphism theorem, the following isomorhpism holds

$$K[t]\,/\,\ker(\mathrm{ev}_\alpha) = K[t]\,/\,(p(t)) \;\cong\; \mathrm{im}(\mathrm{ev}_\alpha) = K[\alpha] \subset F$$

This tells us that $K[\alpha]$ is a subring of F, $K[\alpha]$ is therefore a subring and an integral domain, and ultimately that $(p(t))$ is a prime ideal and that $p(t)$ is an irreducible element (we know that it is nonzero because α is algebraic).

(III) As generators are unique up to multiplication by a unit, we might as well suppose that the polynomial $p(t)$ is monic.

(IV) We lastly need to prove uniqueness. Suppose that there's another irreducible monic polynomial such that $q(t) = 0$. As $q(t) \in \ker(\mathrm{ev}_\alpha) = (p(t))$, $q(t) = p(t) \cdot r(t)$ and as $q(t)$ is irreducible, $r(t)$ has to be a unit. Lastly, $r(t)$ has to be 1 as both $q(t)$ and $p(t)$ are monic polynomials.

$$r(t) \in (K[t])^\times = K^\times = K \qquad \& \qquad p(t) = q(t)$$

$\square$

Lecture 3 (September 9th)

Definition. We let $K[\alpha]$ denote the subring generated by K and α and $K(\alpha)$ denote the subfield generated by K and α as seen below.

$$K[\alpha] = \{f(\alpha) \in F \mid f(t) \in K[t]\}$$

and

$$K(\alpha) = \left\{\frac{f(\alpha)}{g(\alpha)} \in F \mid f(t), g(t) \in K[t]\ g(\alpha) \neq 0\right\}$$

Recall. Take a principle ideal domain (PID) R and a nonzero $p \in R$. The following are equivalent

(i) p is prime

(ii) $(p) \subset R$ is prime

(iii) $(p) \subset R$ is maximal

(iv) p is irreducible

LECTURE 5 (SEPTEMBER 16TH)

Recall. Last time, we have dealt with $K[x]/(p(t))$ and minimal polynomials. Today, we will learn about properties of finite and algebraic extensions.

Definition. (Simple field extensions) A field extension F/K is simple if $F = K(\alpha)$ for some $\alpha \in F$, where $K(\alpha)$ denotes the smallest subfield of F containing $K \cup \{\alpha\}$. Notice that the field extension is constructed from simply adding a single element.

Remark. (Notation) $K(\alpha)$ is the smallest subfield, and $K[\alpha]$ is the smallest subring of F containing $K \cup \{\alpha\}$. Notice that these are equivalent to the definitions introduced last class due to inclusiveness and minimality.

Remark.
(i) $\alpha \in F$ is algebraic over K if and only if

$$\ker\Big(\mathrm{ev}_\alpha : K[t] \to F : f(t) \mapsto f(\alpha)\Big)$$

is nontrivial.

(ii) If α is algebraic over K, then $K[\alpha] = K(\alpha)$. In particular, every element of $K(\alpha)$ can be expressed as a polynomial in α with coefficient in K.

Proof. Consider $\alpha \in F$.

$$\ker(\mathrm{ev}_\alpha) = (p(t))$$

and due to the first isomorphic theorem,

$$K[t]/(p(t)) \cong K[\alpha]$$

If α is algebraic over K, then $(p(t))$ must be a maximal ideal of $K[t]$. This implies that $K[\alpha] = K(\alpha)$. $\square$

Proposition. Let $K \subset K(\alpha) = F$ be a simple extension. Then either

(i) $F \cong K[t]/(p(t))$ for an irreducible monic polynomial $p(t) \in K[t]$ such that $p(\alpha) = 0$

(ii) $F \cong K(t)$, the field of fractions of $K[t]$

In the first case, $[F : K] = \deg(p(t))$. In the second case, $[F : K] = \infty$.

Example. (13.21) One example of a simple extension is

$$\alpha = 2^{1/5} \in \boldsymbol{Q}(2^{1/5})$$

Notice that π is algebraic in $\boldsymbol{R}$ but is not algebraic in $\boldsymbol{Q}$.

Theorem. (7.11) Let K be a field and $f(x)$ be a nonzero polynomial of degree n. Then,

$$K[x]/(f(x))$$

is a n-dimensional vector space over K.

Proof. Notice that the following set is a basis for $K[x]/(f(x))$

$$\{1, x, \dots, x^{n-1}\}$$

□

Remark. (Notation) The minimal polynomial of α over K is denoted $\mathrm{irr}(\alpha, K)$.

13.4 Algebraic Extensions

Definition. (13.30) A field extension F/K is algebraic if every element of F is algebraic over K.

Proposition. (13.31) Every finite extension F/K is algebraic.

Proof. Fix an element $\alpha \in F$. Set $\dim_K F = n$. Then

$$\{1, \alpha, \alpha^2, \dots, \alpha^n\}$$

is linearly dependent over K, that is, $a_0 + a_1\alpha + \dots + a_n\alpha^n = 0$ for some $(a_0, \dots, a_n) \neq \mathbf{0} \in K^{n+1}$. Take

$$f(t) = a_0 + a_1 t + \dots + a_n t^n$$

in $K[t]$ and it is a polynomial with coefficients in K such that $f(\alpha) = 0$. □

Proposition. (13.13) Let $K \subset E \subset F$ be field extensions. If E/K and F/E are both finite, then F/K is also finite. Furthermore,

$$[F : K] = [F : E][E : K]$$

Proof. Let $\{f_i\}_{i=1}^n$ be an E-basis of F and $\{e_j\}_{j=1}^n$ be a K-basis of E. Then $\{f_i e_j\}$ is a K-basis of F. The rest is for you! □

Remark. If $\alpha \in F$ is algebraic over K, then $K(\alpha)/K$ is finite, hence algebraic. This is because

$$\dim_K K(\alpha) = \deg\Big(\operatorname{irr}(\alpha, K)\Big)$$

Here,

$$\dim_K K(\alpha) = \dim_K K[t]/(\operatorname{irr}(\alpha, K))$$

as

$$K[t]/(\operatorname{irr}(\alpha, K)) \cong K[\alpha] = K(\alpha)$$

due to the 3rd isomorphism theorem.

Proposition. A field extension $F\,/\,K$ is finite if and only if there are algebraic elements $\alpha_1, \alpha_2, \ldots, \alpha_r$ over K such that $F = K(\alpha_1, \alpha_2, \ldots, \alpha_r)$.

$$\begin{tikzcd}
F \arrow[dd, dash] \arrow[dr, dash] & \\
& K(\alpha_1, \alpha_2, \ldots, \alpha_n) \arrow[dl, dash] \\
K &
\end{tikzcd}$$

Proof. (if) Suppose we are given algebraic elements $\alpha_1, \ldots, \alpha_r$ over K such that $F = K(\alpha_1, \ldots, \alpha_r)$. Consider the tower of finite towers

$$\begin{tikzcd}
F = K(\alpha_1, \ldots, \alpha_r) \arrow[d, dash] \\
(K(\alpha_1, \alpha_2))(\alpha_3) \arrow[d, dash] \\
(K(\alpha_1))(\alpha_2) = K(\alpha_1, \alpha_2) \arrow[d, dash] \\
K(\alpha_1) \arrow[d, dash] \\
K
\end{tikzcd}$$

$\square$

Lecture 6 (September 18th)

Remark. Las time, we learned about a tower of finite extensions. Today, we will learn about towers of algebraic extensions and constructability.

Proposition. If $\alpha \in F$ is algebraic over K, then

$$[K(\alpha) : K] = \deg(\mathrm{irr}(\alpha, K))$$

Proposition. (13.13) If $K \subset E$ and $E \subset F$ are finite extensions, then so is $K \subset F$. Furthermore,

$$[F : K] = [F : E][E : K]$$

Remark. The converse is also true.

Proposition. An extension $F \,/\, K$ is finite if and only if there are algebraic elements $\alpha_1, \ldots, \alpha_r$ over K such that $F = K(\alpha_1, \ldots, \alpha_r)$.

Proof. (only if) Assume that $F \,/\, K$ is finite. Then $F = K(\alpha_1, \ldots, \alpha_r)$ where $\{\alpha_1, \ldots, \alpha_k\}$ is a K-basis of F.

(if) If $r = 1$, then $K(\alpha_1) \,/\, K$ is finite by the proposition above. Inductively, we may assume that $r = 2$. In this case, consider the tower

$$\begin{array}{c} K(\alpha_1, \alpha_2) \\ | \\ K(\alpha_1) \\ | \\ K \end{array}$$

Since α_2 is algebraic over $K(\alpha_1)$, it follows from the above proposition that $K(\alpha_1, \alpha_2) = \Big(K(\alpha_1)\Big)(\alpha_2)$ is finite over $K(\alpha_1)$. Using proposition (13.13), we see that $K(\alpha_1, \alpha_2) \,/\, K$ is finite ($K(\alpha_1) \,/\, K$ is finite by the proposition above). $\square$

Proposition. (13.35) If $K \subset E$ and $E \subset F$ are field extensions, so is $K \subset F$.

Remark. The converse is also true.

$$\begin{array}{c} \alpha \in F \\ | \\ E \\ | \\ K \end{array}$$

Proof. Let $\alpha \in F$ be an element. Then $f(\alpha) = 0$ for some non-zero polynomial

$$f(t) = e_0 + e_1 t + e_2 t^2 + \ldots + e_n t^n$$

where $e_i \in E$ for all i. Consider a tower

$$\begin{tikzcd} F \arrow[d, dash] \arrow[dr, dash] & \\ E \arrow[r, dash] \arrow[d, dash] & K(e_0, e_1, \ldots, e_n) \arrow[dl, dash] \\ K & \end{tikzcd}$$

Note that α is algebraic over $K(e_0, e_1, \ldots, e_n)$ so that $K(e_0, e_1, \ldots, e_n, \alpha) \,/\, K(e_1, \ldots, e_n)$ is finite. Since $E \,/\, K$ is algebraic, it follows from proposition (the proposition tha tif $F \,/\, K$ is finite, $F = K(\alpha_1, \ldots, \alpha_r)$ for algebraic elements α_i) that $K(e_0, \ldots, e_n) \,/\, K$ is finite, hence algebraic. In particular, α is algebraic over K. ☐

Proposition. (13.33) Let $E \subset F$ be the subset consisting of algebraic elements over K. Then E is a subfield of F.

Proof. Let α and β be algebraic elements over K. We show that $\alpha \pm \beta$, $\alpha\beta$, and α/β if $\beta \neq 0$ are all algebraic over K. Do this yourself using the above! ☐

13.5 Applications: Geometric Impossibilies

Remark. Drawing shapes only using a straightedge and a compass. We know that two possible shapes are an equalateral triangle and a line perpendicular to the line passing through the middle point. What about the following?

(i) The side of a square with the same area as a circle

(ii) The side of a cube whose volume is 2

(iii) A line trisecting the angle formed by two lines

Definition. (13.36) A real number is constructable if it is a coordinate of a point one can construct by a straightedge and a compass:

(i) Draw a line through two points you have already constructed

(ii) Draw a circle with the center at a point you have constructed through another point you have constructed

(iii) Mark a point of interesection of two lines, a line and a circle, or two circles you have already constructed

Lecture 7 (September 23rd)

Remark. Last time we talked about constructible numbers. This class, we address some old questions on constructibility. Recall that we only allowed a straightedge (a ruler without marks) and compasses.

Definition. (13.36) A constructible number is obtained from the following operations:

(i) Making a straightline between two points

(ii) Drawing a circle around a point

(iii) Making interactions between lines and circles

Example. $\frac{1}{2}$ and $\frac{\sqrt{3}}{2}$ are constructible. In fact, all rational numbers are constructible. How about

(i) Squaring a circle, or $\sqrt{\pi}$

(ii) Doubling the cuble, or π

(iii) Trisecting the angle $60°$, or $20°$

Remark. If α and β are constructible real numbers, then so are $\alpha \pm \beta$, $\alpha\beta$, and α/β (if $\beta \neq 0$). Consequently, the set of constructible numbers is a subfield of $\boldsymbol{R}$ containing $\boldsymbol{Q}$.

Remark. If α is constructible, then so is its square root $\sqrt{\alpha}$.

Theorem. (13.38) Let α be a constructible number. Then $\alpha \in F$, where F is a field extension of $\boldsymbol{Q}$ such that $[F : Q]$ is a finite power of 2. In particular, α is an algebraic number and $[\boldsymbol{Q}(\alpha) : \boldsymbol{Q}]$ is a finite power of 2.

Proof. It suffices to show the following – in any stage of a construction, if all the constructed numbers lie in a field E, then applying one of the basic operations the same data lies in a field $E(u)$, where u is algebraic of degree at most over E.

Note that at any stage of a construction, we consider:

(i) A coordinate of points that have been constructed.

(ii) A coefficient in the equations of the lines and circles that have been constructed.

It is enough to intersect a circle and a line to get new constructible points. For this, we need to solve

$$\begin{cases} x^2 + y^2 + ax + by + c = 0 \\ y = mx + r \end{cases}$$

where $a, b, c \in E$ and $m, r \in E$. Notice that

$$x = \frac{-(mr + bm + a) \pm \sqrt{(2mr + bm + a)^2 - 4(m^2 + 1)(r^2 + br + c)}}{2(m^2 + 1)}$$

which is in $E(u)$ and $u = \Delta$. $\square$

Corollary. (13.40) The three inquries from above are impossible. That is, none of the numbers $\sqrt{\pi}$, $2^{1/3}$, and $\cos 20°$ are constructible.

Proof. (I) If $\sqrt{\pi}$ is constructible, then so is π. However, as we expect all constructible numbers to be algebraic, and π is strictly not over $\boldsymbol{Q}$, the claim is false.

(II) If $2^{1/3} = \alpha$, the irreducible polynomial is $t^3 - 2 \in \boldsymbol{Q}[t]$. However,

$$\begin{aligned} [\boldsymbol{Q}(\alpha) : \boldsymbol{Q}] &= \deg(\operatorname{irr}(\alpha, \boldsymbol{Q})) \\ &= \deg(t^3 - 2) \\ &= 3 \end{aligned}$$

which is not a finite power of 2.

(III) For $\cos 20° = \alpha$, note that α is a zero of $8t^3 - 6t - 1 \in \boldsymbol{Q}[t]$. Again, the root test gureentees that the polynomial is irreducible over $\boldsymbol{Q}$. Same as the above, the degree of the polynomial is not a finite power of 2, and the number is not construtible. $\square$

Lecture 8 (September 25th)

Remark. Last time, we've seen how $[\boldsymbol{Q}(\alpha) : \boldsymbol{Q}]$ is a finite power of 2 for a constructible α. Today, we'll learn about splitting fields.

Theorem. (13.38) Let α be a constructible number. Then,

(i) $\alpha \in F$, where $[F : \boldsymbol{Q}]$ is a finite power of 2

(ii) α is algebraic over $\boldsymbol{Q}$

(iii) $[\boldsymbol{Q}(\alpha), \boldsymbol{Q}]$ is again a finite power of 2

Theorem. To prove this theorem, it is enough to show that: $C = \{\text{numbers that have been constructed}\} \subset E$, a field, acted on by the basic operators gives $C \cup \{\text{newly constructed numbers}\} \subset E(u)$ where $[E(u) : E] \leq 2$. For example,

$$\{y = mx + r\} \cap \{x^2 + y^2 + ax + by + c = 0\}$$

solves to give

$$x' = \frac{-(2mr + bm + a \pm \sqrt{(2mr + bm + a)^2 - 4(m^2 + 1)(r^2 + br + c)})}{2(m^2 + 1)}$$

By setting $u = \sqrt{(2mr + bm + a)^2 - 4(m^2 + 1)(r^2 + br + c)}$, we find that both x' and $y' = mx' + r$ lies in $E(u)$. It is trivial from here that the field extension has a dimension lower than or equal to 2.

Chapter 14. Normal and Separable Extensions, and Splitting Fields

Theorem. (Kronecker) Let K be a field, and let $p(t) \in K[t]$ be irreducible. Then there exists an extension field that contains a zero α of $p(t)$.

Proof. Set

$$L = K[t]/(p(t))$$

and we have a field extension that contains a root of $p(t)$. Let $\alpha = t + (p(t)) \in L$. Observe that

$$p(t + (p(t))) = p(t) + (p(t)) = (p(t)) = 0$$

and that α is a root of the polynomial $p(t)$ in L. It is always great to gain mastery in multiple notations. We can also write the above using 'bar' notation as

$$\overline{p}(\alpha) = \overline{p}(\overline{t}) = \overline{p(t)} = 0$$

$\square$

Example. Let $K = \boldsymbol{R}$ and $p(t) = t^2 + 1 \in \boldsymbol{R}[t]$. The field of complex numbers can be defined as

$$\boldsymbol{C} = \boldsymbol{R}[t]/(t^2 + 1)$$

and would contain the root $\alpha = \overline{t}$. Note that

(i)

$$\overline{t}^2 + \overline{1} = \overline{t^2 + 1} = \overline{0} \in \boldsymbol{R}[t]/(t^2 + 1)$$

That is, $\bar{t}$ is a zero of
$$x^2 + 1 \in \Big(\boldsymbol{R}[t]/(t^2+1)\Big)[x]$$

(ii) Note that the map $\iota : \boldsymbol{R} \hookrightarrow \boldsymbol{R}[t]/(t^2+1)$ is injective

(iii) $\boldsymbol{R}[t]/(t^2+1)$ is a field since $\boldsymbol{R}[t]$ is a PID and t^2+1 is irreducible in $\boldsymbol{R}$.

Lecture 9 (September 30th)

Definition. (14.2) Let $f(t) \in K[t]$. A splitting field for $f(t)$ is a field extension F/K such that $f(t)$ factors into linear terms
$$c \cdot \prod_{i=1}^{n} (t - \alpha_i)$$
with $F = K(\alpha_1, \alpha_2, \ldots, \alpha_n)$.

Definition. (13.19) A field F is algebraically closed if every nonconstant polynomial $f(t) \in F[t]$ has a zero in F.

Remark. If $K = \boldsymbol{Q}$, the existence of a the splitting field of $f(t)$ is immediate: if $f(t) = c \cdot \prod_{i=1}^{n}(t - \alpha_i)$ in $\boldsymbol{C}[t]$, then
$$F = K(\alpha_1, \ldots, \alpha_n) \subset \boldsymbol{C}$$

Theorem. (14.6) There exists a splitting field of $f(t) \in K[t]$.

Proof. We prove this theorem inductively. Assume $\deg(f) = 1$. We simply take $L = K$ and L/K is a field extension such that $f(t)$ factors into linear terms.

Assume that the statement holds for all polynomials with $\deg(f) < n$. For a polynomial $f(t)$ of degree n, we can pull out an irreducible factor $p(t)$ and using Kronecker's theorem, we have a field K_1 such that $f(t) = (t-\alpha)^m f_1(t)$ in K_1 with $\deg f_1(t) < n$. By assumption, we can find a splitting field of K_1 where $f_1(t)$ splits into linear factors:
$$f(t) = (t-\alpha)^m \cdot c \cdot \prod_{i=1}^{n-1} (t - \alpha_i) \in L$$
and L/K is a field extension such that $f(t)$ factors into linear terms with $L = K(\alpha_1, \alpha_2, \ldots, \alpha_n)$. $\square$

Theorem. (14.3) Let K be a field. Then there exists an algebraic extension $\bar{K}/K$ such that $\bar{K}$ is algebraically closed.

$$\begin{tikzcd}
\bar{K} \arrow[dd, dash] \arrow[dr, dash] & \\
 & L \arrow[dl, dash] \\
K &
\end{tikzcd}$$

Remark. Such an extension is unique: if F and L are two such extensions of K, then there exists an isomorphism $\sigma : F \to L$ such that $\sigma|_K = \mathrm{id}_K$ (that is, $\sigma(a) = a$ for all $a \in K$).

$$\begin{tikzcd}
F \arrow[r, "\sigma"] & L \\
K \arrow[u] \arrow[r, "\mathbf{1}"] \arrow[ur, phantom, "\circlearrowright"] & K \arrow[u]
\end{tikzcd}$$

Definition. (14.4) We call $\bar{K}$ the algebraic closure of K.

Lecture 10 (October 2nd)

Remark. Last class we talked about algebraically closed fields. This class we will be talking about isomorphism extensions theorem.

Definition. (13.19) A field K is algebraically closed if for every algebraic extension F of K, $F = K$, or equivalently, every nonconstant polynomial $f(t) \in K[t]$ splits into linear factors.

Definition. (14.4) $\bar{K}$ is an algebraic closure of K if

(i) $\bar{K}/K$ is algebraic

(ii) $\bar{K}$ is algebraically closed

Theorem. (Isomorphic extension) Let $\sigma : K_1 \to K_2$ be an isomorphism of fields. Let F/K_1 be an algebraic extension. Then there exists a field embedding $\bar{\sigma} : F \to \bar{K}_2$ extending σ (that is, $\bar{\sigma}|_{K_1} = \sigma$).

$$
\begin{tikzcd}
 & \bar{K}_2 \\
F \arrow[ru, "\bar{\sigma}"] & \\
K_1 \arrow[u] \arrow[r, "\sigma"] \arrow[ur, phantom, "\circlearrowright"] & K_2 \arrow[uu]
\end{tikzcd}
$$

Proposition. (14.1) Let F/K be a field extension and let $p(x) \in K[x]$ be irreducible. Then there is a 1-1 correspondence between the sets of zeros of $p(t)$ in F and the set of K-embeddings $K[t]/(p(t)) \to F$ which extends the inclusion $K \hookrightarrow F$.

Proof. Suppose that there is a K-embedding $\sigma : K[t]/(p(t)) \to F$. $\sigma(\bar{t}) \in F$ and $\sigma(p(\bar{t})) = p(\sigma(\bar{t})) = 0$, such that $\sigma(\bar{t})$ is a root of the polynomial $p(t)$ in F.

Conversely, let α be a zero of $p(t)$ in F. $\mathrm{ev}_\alpha : K[t] \to F$ satisfies $\ker(\mathrm{ev}_\alpha) = (p(t))$ and there exists $\sigma : K[t]/(p(t)) \to F$ by the 1st isomorphism theorem. □

Remark. We now write $\sigma : R_1 \to R_2$ for a ring homomorphism. If $f(t) \in R_1[t]$ is a polynomial, we let $f_\sigma(t)$ denote the image of $f(t)$ under the induced homomorphism $R_1[t] \to R_2[t]$. Concretely, for $f(t) = \sum_{i=0}^n a_i t^i$ we have $f_\sigma(t) = \sum_{i=0}^n \sigma(a_i) t^i$.

Proposition. Let $\sigma : K_1 \to K_2$ be an isomorphism of fields. Let F_1/K_1 and F_2/K_2 be field extensions. Let $p(t)$ be an irreducible polynomial in $K_1[t]$. Suppose that $\alpha \in F_1$ is a zero of $p(t)$ and that $\beta \in F_2$ is a zero of $p_\sigma(t)$. Then there exists a unique isomorphism $\bar{\sigma} : K_1(\alpha) \to K_2(\beta)$ such that $\bar{\sigma}|_{K_1} = \sigma$ and that $\bar{\sigma}(\alpha) = \beta$.

$$
\begin{tikzcd}
F_1 & F_2 \\
K_1(\alpha) \arrow[u] \arrow[r, "\bar{\sigma}"] & K_2(\beta) \arrow[u] \\
K_1 \arrow[u] \arrow[r, "\sigma"] \arrow[ur, phantom, "\circlearrowright"] & K_2 \arrow[u]
\end{tikzcd}
$$

Again, we have

$$
\begin{cases} p(t) \in K_1[t] \\ p(\alpha) = 0 \end{cases} \qquad \& \qquad \begin{cases} p_\sigma(t) \in K_2[t] \\ p_\sigma(\beta) = 0 \end{cases}
$$

Proof. (I) For isomorphism $\sigma = \mathbf{1} : K \to K$, we can simply consider a irreducible polynomial such that $p(\alpha) = p(\beta) = 0$. Then, we define $\bar{\mathbf{1}} : K(\alpha) \to K(\beta)$ by sending $f(\alpha)/g(\alpha) \in K(\alpha)$ to $f(\beta)/g(\beta) \in K(\beta)$.

(II) More concretely, define the function $\bar{\mathbf{1}} : K(\alpha) \to K(\beta)$ to be defined through the following diagram,

$$
\begin{tikzcd}
 & F_1 & F_2 & \\
K[t]/(p(t)) \arrow[r, dash, "\bar{\mathrm{ev}}_\alpha"] \arrow[rrr, bend left, "\mathbf{1}"] & K(\alpha) \arrow[u] \arrow[r, "\bar{\mathbf{1}}"] & K(\beta) \arrow[u] \arrow[r, dash, "\bar{\mathrm{ev}}_\beta"] & K[t]/(p(t)) \\
 & K \arrow[u] \arrow[r, "\mathbf{1}"] & K \arrow[u] &
\end{tikzcd}
$$

as

$$\bar{\mathrm{ev}}_\beta \circ (\bar{\mathrm{ev}}_\alpha)^{-1}$$

□

Lecture 11 (October 14th)

Remark. A (long long long)^3 time ago, we learned about the existence of spliiting fields and isomorphism extensions. Today, we learn about how splitting fields are unique.

Proposition. (14.1) Let $K \subset F$ be a field extension and $p(t) \in K[t]$ be an irreducible element. There is a one-to-one correspondence between the set

$$\{\alpha \in F \mid p(\alpha) = 0\}$$

and

$$\{\sigma : K[t]/(p(t)) \to F \mid \sigma|_K = 1_K\}$$

Theorem. (Isomorphism extension for simple algebraic extensions)

$$
\begin{tikzcd}
 & E & F & \\
K[t]/(p(t)) \arrow[rrr, bend left, "\mathbf{1}"] & K(\alpha) \arrow[l, "\cong"'] \arrow[u] \arrow[r, "\cong"] & K(\beta) \arrow[u] \arrow[r, "\cong"] & K[t]/(p(t)) \\
 & K \arrow[u] \arrow[r, "\mathbf{1}"] & K \arrow[u] &
\end{tikzcd}
$$

Let $\alpha \in E$ be algebraic over K and $\beta \in F$ be a zero of $p(t) = \operatorname{irr}(\alpha, K)$. Then, there exists an isomorphism $\sigma : K(\alpha) \to K(\beta)$ such that $\sigma|_K = 1_K$

Theorem. (Isomorphism extension for finite extensions) Let E/K be a finite extension. Fix an algebraic closure $\bar{K}$ of K. Then there exists a K-embedding $\sigma : E \to \bar{K}$.

$$
\begin{tikzcd}
E & \bar{K} \\
\big(K(\alpha_1)\big)(\alpha_2) \arrow[u] \arrow[r, "\sigma_2"] & \big(K(\beta_1)\big)(\beta_2) \arrow[u] \\
K(\alpha_1) \arrow[u] \arrow[r, "\sigma_1"] & K(\beta_1) \arrow[u] \\
K \arrow[u] \arrow[r, "\mathbf{1}"] & K \arrow[u]
\end{tikzcd}
$$

Proof. Since E/K is finite, $E = K(\alpha_1, \alpha_2, \ldots, \alpha_n)$ for some algebraic elements $\alpha_1, \ldots, \alpha_n$ over K. Using this fact and induction, we can prove our theorem.

For $\alpha \in E$ which is algebraic in K, we can set $p_1(t) = \operatorname{irr}(\alpha, K) \in K[t]$. Let $\beta_1 \in \bar{K}$, a zero of $p(t)$ in $\bar{K}$. Then, let $\alpha_2 \in E$, which is again algebraic in $K(\alpha_1)$. We can set $p_2(t) = \operatorname{irr}(\alpha_2, K(\alpha_1)) \in \big(K(\alpha_1)\big)[t]$. We can choose a zero $\beta_2 \in \bar{K}$ of $p_2(t) \in \big(K(\beta_1))\big)[t]$. $\square$

Remark. Let $p(t) \in K[t] \subset \bar{K}[t]$ be an irreducible element and

$$p(t) = (t - \alpha_1)^{m_1}(t - \alpha_2)^{m_2} \ldots (t - \alpha_n)^{m_n}$$

is $m_i = m_j$ for $i \neq j$?

Theorem. (Isomorphism extension theorem) Let K be a field, and let $f(t) \in K[t]$ be a nonconstant polynomial. Suppose that F_1/K and F_2/K are splitting fields for $f(t)$. Then there exists a K-isomorphism $\sigma : F_1 \to F_2$.

Proof. (i) Choose a zero α_1 of $f(t)$ in F_1 and let $p(t) = \mathrm{irr}(\alpha_1, K)$

(ii) Choose a zero β_1 of $p(t)$ in F_2

We then have an isomorphism σ_1 between $K(\alpha_1)$ and $K(\beta_1)$. We can continue and achieve:

$$\begin{tikzcd}
F_1 & F_2 \\
\big(K(\alpha_1)\big)(\alpha_2) \arrow[u] \arrow[r, "\sigma_2"] & \big(K(\beta_1)\big)(\beta_2) \arrow[u] \\
K(\alpha_1) \arrow[u] \arrow[r, "\sigma_1"] & K(\beta_1) \arrow[u] \\
K \arrow[u] \arrow[r, "\mathbf{1}"] & K \arrow[u]
\end{tikzcd}$$

□

Remark. Let $f(t) = (t - \alpha_1)\dots(t - \lambda_n)$ in $\bar{K}[t]$. Then there exists a K-isomorphism $\sigma_i : F_1 \to K(\lambda_1, \dots, \lambda_n) \subset \bar{K}$.

Lecture 12 (October 16th)

Remark. Last time we learned about extended isomorphisms. Today, we will look at some specific examples and separable extensions.

Proposition. (14.1) Let $p(t) \in K[t]$ be a irreducible polynomial and F/K be an extension of fields. $\{$zeros of $p(t)$ in $F\}$ is isomorphic to $\{K$-embeddings $K[t]/(p(t)) \to F\}$.

Theorem. (Isomorphism extension theorem) Let α be a zero of an irreducible polynomial $p(t) \in K_1[t]$ and β be a zero of $p_\sigma(t) \in K_2[t]$.

$$\begin{tikzcd}
F_1 & F_2 \\
K_1(\alpha) \arrow[u] \arrow[r, "\exists! \cong"] & K_2(\beta) \arrow[u] \\
K_1 \arrow[u] \arrow[r, "\cong"] & K_2 \arrow[u]
\end{tikzcd}$$

One of the concequences of the theorem is the uniqueness of splitting fields: for any non-constant $f(t) \in K[t]$ and splitting fields F_1 and F_2 for $f(t)$, there exists a K-isomorphism

$$\begin{tikzcd}
F_1 \arrow[rr, "\cong"] \arrow[dr] & & F_2 \\
& K \arrow[ur] &
\end{tikzcd}$$

Example. (14.13) Let $p(t) = 1 + t + t^2 + t^3 + t^4 \in \boldsymbol{Q}[t]$.

(i) The splitting field for $p(t)$ is $\boldsymbol{Q}(\xi)$

(ii) $[E : \boldsymbol{Q}] = 4$

(iii) The number of $\boldsymbol{Q}$-embeddings

$$E \to \bar{\boldsymbol{Q}}$$

is 4

Proof. The (i), note that $(t-1)p(t) = t^5 - 1$. Fix a 5th primitive root of unity (that is, a generator of the multiplicative group $\{z \in \boldsymbol{C} \mid z^5 - 1 = 0\}$ which is cyclic). Set $\xi = \exp(2\pi i/5)$ so that $\xi, \ldots, \xi^4$ are roots of $p(t)$ in $\bar{\boldsymbol{Q}}$. Therefore,

$$E = \boldsymbol{Q}(\xi, \xi^2, \xi^3, \xi^4) = \boldsymbol{Q}(\xi)$$

For (ii), $[E : \boldsymbol{Q}] = [\boldsymbol{Q}(\xi) : \boldsymbol{Q}] = \deg(\mathrm{irr}(\xi, \boldsymbol{Q})) = \deg(p(t)) = 4$ by Eisenstein's criterion, which shows that $p(t)$ is irreducible over $\boldsymbol{Q}$.

For (iii), we first observe that $E = \boldsymbol{Q}(\xi)$ is isomorphic to $\boldsymbol{Q}[t]/(p(t))$. Applying proposition 14.1, the number of $\boldsymbol{Q}$-embeddings from $E = \boldsymbol{Q}(\xi) \to \bar{\boldsymbol{Q}}$ equals the number of distinct zeros of $p(t)$ in $\bar{\boldsymbol{Q}}$, which is 4. $\square$

Example. (14.14) Let $p(t) = t^3 - 2 \in \boldsymbol{Q}[t]$.

(i) The splitting field E for $p(t)$ is $\boldsymbol{Q}(\alpha, \xi)$.

(ii) $[E : \boldsymbol{Q}]$ is 3.

(iii) The number of $\boldsymbol{Q}$-embeddings

$$E \to \bar{\boldsymbol{Q}}$$

is 3.

Proof. For (i), fix a 3rd primitive root of unity ξ. For example, $\xi = \exp(2\pi i/3)$. Then the zeros of $t^3 - 2$ in $\boldsymbol{C}$ (and in $\bar{\boldsymbol{Q}}$) are $\alpha, \alpha\xi$, and $\alpha\xi^2$, where $\alpha = 2^{1/3}$. This implies that

$$E = \boldsymbol{Q}(\alpha, \alpha\xi, \alpha\xi^2) = \boldsymbol{Q}(\alpha, \xi)$$

For (ii), $[E, \boldsymbol{Q}] = [\boldsymbol{Q}(\xi, \alpha) : \boldsymbol{Q}] =$ as

$$\begin{tikzcd}
\big(\boldsymbol{Q}(\xi)\big)(\alpha) \arrow[d, dash] \\
\boldsymbol{Q}(\xi) \arrow[d, dash] \\
\boldsymbol{Q}
\end{tikzcd}$$

and

$$\begin{aligned}
[\boldsymbol{Q}(\xi, \alpha), \boldsymbol{Q}] =& [\boldsymbol{Q}(\xi, \alpha) : \boldsymbol{Q}(\xi)][\boldsymbol{Q}(\xi) : \boldsymbol{Q}] \\
=& \deg(\mathrm{irr}(\alpha, \boldsymbol{Q}(\xi))) \cdot \deg(\mathrm{irr}(\xi, \boldsymbol{Q})) \\
=& \deg(t^3 - 2) \cdot \deg(t^2 + t + 1) = 3 \cdot 2 = 6
\end{aligned}$$

For (iii),

$$\begin{tikzcd}
 & \bar{\boldsymbol{Q}} \arrow[d, dash] \\
\boldsymbol{Q}(\alpha, \xi) \arrow[r, "3"] \arrow[d, dash] & \boldsymbol{Q}(\xi^2) \arrow[d, dash] \\
\boldsymbol{Q}(\xi) \arrow[r, "2"] \arrow[d, dash] & (*) \arrow[d, dash] \\
\boldsymbol{Q} \arrow[r] & \boldsymbol{Q}
\end{tikzcd}$$

Observation is that if E/K and F/K are field extensions, $\sigma : E \to F$, a K-embedding $\alpha \in E$, algebraic over K implies that $\sigma(\alpha)$ is a zero of $\mathrm{irr}(\alpha, K)$. We make a certain remark. How many $\boldsymbol{Q}$-automorhpisms on $\boldsymbol{Q}(\xi)$ are there? $\square$

Lecture 13 (October 28th)

Recall. Let $f(t) \in K[t]$ and E be the splitting field for $f(t)$. We have seen (1) $[E : K]$ and (2) the number of K-embeddings $E \to \bar{K}$. Today's goal is to study (2) and separability.

Chapter 14.4 Separable Extensions

Definition. (14.9) Let K be a field. A polynomial $f(t) \in K[t]$ is separable if its factorisation over $\bar{K}$ has no multiple roots. Let F/K be a algebraic extension. An element $\alpha \in F$ is separable over K if the minimal polynomial $\mathrm{irr}(\alpha, K)$ over K is separable. We say that F/K is separable if every $\alpha \in F$ is separable over K.

Proposition. Let $f(t) \in K[t]$ be a nonzero polynomial. Then, $f(t)$ is separable over K if $f(t)$ has no multiple roots in $\bar{K}$. The following are equivalent:

(i) $f(t)$ is separable over K

(ii) $f(t)$ and $f'(t)$ have no common root in $\bar{K}$

(iii) $\gcd(f(t), f'(t)) = 1$

Proof. The proof is yours! □

Proposition. (14.23) Let K be a field of characteristic zero and let $f(t) \in K[t]$ be a irreducible polynomial. Then, $f(t)$ is separable. Moreover, algebraic extensions of fields of characteristic zero are separable.

Proof. Set $f(t) = a_n t^n + \ldots + a_1 t + a_0$ so that $f'(t) = n a_n t^{n-1} + \ldots + a_1$. Note that $\deg(f') < \deg(f)$ since $\mathrm{char}(K) = 0$. Suppose that $\alpha \in \bar{K}$ is a common root of $f(t)$ and $f'(t)$. Then $\mathrm{irr}(\alpha, K) | f'(t), f(t)$. Since $f(t)$ is irreducible over K, $\deg\big(\mathrm{irr}(\alpha, K)\big) = \deg\big(f(t)\big)$. This contradicts to the observation that $\mathrm{irr}(\alpha, K) | f'(t)$ because $\deg(f') < \deg(f)$. □

Proposition. (14.24) (Primative element theorem) Every finite separable extension is simple.

Proof. The proof is ours and is technical. □

Corollary. (14.25) Let F/K be a finite separable extension. Then there are exactly $[F : K]$ number of K-embeddings $F \to \bar{K}$.

Proof. By the primative element theorem, $F = K(\alpha)$ for some $\alpha \in F$. Since α is separable over K, $\operatorname{irr}(\alpha, K)$ has no multiple roots in $\bar{K}$. Applying proposition (14.3), the number of K-embeddings $F = K(\alpha) \hookrightarrow \bar{K}$ equals the number of distinct roots in $\bar{K}$ of $\operatorname{irr}(\alpha, K)$ which is equal to

$$\deg\bigl(\operatorname{irr}(\alpha, K)\bigr) = [K(\alpha) : K]$$

$\square$

Remark. Let $\alpha \in F$ be an algebraic element over K. Then the number of K-embeddings $K(\alpha) \hookrightarrow \bar{K}$ is the number of distinct zeros of $\operatorname{irr}(\alpha, K)$ in $\bar{K}$, and both are less or equal to $\deg\bigl(\operatorname{irr}(\alpha, K)\bigr) = [K(\alpha), K]$. According to corollary (14.25), the euqality holds if and only if α is separable over K.

Definition. (14.26) The separability degree of an algebraic extension F/K denoted $[F : K]_s$, is the number of K-embeddings $F \hookrightarrow \bar{K}$.

Lecture 14 (October 30th)

Remark. Last class, we learned about separable extensions, and today we will be learning about separability degree.

Definition. (14.9) We called a $\alpha \in F$ separable if $\operatorname{irr}(\alpha, K)$ has no repeated roots in $\bar{K}$.

Proposition. (14.23) If $\operatorname{char}(K) = 0$, every irreducible polynomial in $K[x]$ is separable.

Proposition. (14.24) (Prime element theorem) Finite separable extensions are simple.

Corollary. (14.25) For a finite separable extension F/K, the number of K-embeddings $F \to \langle K$ is $[F : K]$. In fact, for an algebraic $\alpha \in F$ over K,

$$|\{F \overset{\sigma}{\to} \bar{K} \mid \sigma|_K = 1_K\}| = |\{\beta \in \bar{K} \mid (\operatorname{irr}(\alpha, K))(\beta) = 0\}| \leq \deg(\operatorname{irr}(\alpha, K)) = [K(\alpha), K]$$

Definition. (14.26) Let F/K be an algebraic extension. Then

$$[F : K]_s = |\{\sigma : F \to \bar{K} \mid \sigma|_K = 1_K\}$$

where $[F : K]_s$ denotes the separability degree of F/K.

Remark. Definition (14.26) is independent of the choice of the algebraic closure of $\bar{K}$.

Proposition. (14.28) If F/E and E/K are algebraic extensions then

$$[F : K]_s = [F : E]_s [E : K]_s$$

Proof. Proof will be further inquired in the supplementary material. $\square$

Theorem. Let F/K be a finite extension. Then F/K is separable if and only if $[F : K]_s = [F : K]$.

Remark. In the case where $F = K(\alpha)$, we already have observed the result.

Remark. For a finite extension F/K,

$$[F : K]_s \leq [F : K]$$

Proof. (if) Let $\alpha \in F$. Consider the tower $F/K(\alpha)/K$. Note that

$$\begin{aligned}[F : K] =&[F : K(\alpha)][K(\alpha) : K] \\ [F : K]_s =&[F : K(\alpha)]_s[K(\alpha) : K]_s\end{aligned}$$

and $[F : K]$ implies that, through the remark, $[F : K(\alpha)] = [F : K(\alpha)]_s$ and $[K(\alpha) : K] = [K(\alpha) : K]_s$. That is, α is separable over K.

(only if) Ours! $\square$

Corollary. Let $\alpha_1, \alpha_2, \ldots, \alpha_n$ be elements of $\bar{K}$. Then $\alpha_1, \ldots, \alpha_n$ are separable over K if and only if $K(\alpha_1, \ldots, \alpha_n)$ is separable over K.

Corollary. Let F/E and E/K be finite extensions. F/K is separable if and only if F/E and E/K are separable.

Proof. To prove this, go back to an analogous statement for algebraic extensions. $\square$

Corollary. Let F/K be an algebraic extension then the set of separable elements of F over K forms a subfield of F.

Lecture 15 (November 4th)

Remark. Last time, we saw various properties of separable extensions. Today, we will learn about normal extensions.

Proposition. (14.23) If $\mathrm{char}(K) = 0$, then every algebraic extension of K is separable.

Proposition. (14.24) (Primitive element theorem) Every finite separable extension is simple.

Definition. (14.26) Let F/K be an algebraic extension. $[F : K]_s$ denotes the separability degree of F/K which is equal to the number of K-embeddings from $F \to \bar{K}$.

Proposition. (14.28) Let $F/E/K$ be algebraic extensions.

$$[F : K]_s = [F : E]_s[E : K]_s$$

Corollary. If F/K is finite, $[F : K]_s < \infty$ (and $[F : K]_s$ divides $[F : K]$).

Chapter 14.3 Normal Extensions

Definition. (14.15) An algebraic extension F/K is normal if for every $\alpha \in F$, the minimal polynomial $\mathrm{irr}(\alpha, K)$ splits in F.

Remark. F/K is normal if and only if the following condition holds: if F contains a zero of an irreducible polynomial $p(t) \in L[t]$, then $p(t)$ splits into linear factors over F.

Remark. (Motivation) Let $p(t) \in K[t]$ be irreducible. Say $p(t) = \prod_{i=1}^{n}(t - \lambda_i) \in \bar{K}[t]$. Consider a K-embedding $\sigma : E \hookrightarrow \bar{K}$ where $E = K(\lambda_1, \ldots, \lambda_n)$ is the splitting field of $p(t)$. We want to figure out the image $\sigma(E) \subset \bar{K}$. We claim that $\sigma(E) = E$. For this, observe that the image of λ_i under σ is α_j for some j. In other words, σ permutes $\{\lambda_1, \lambda_2, \ldots, \lambda_n\}$ ($\sigma : \{\lambda_1, \ldots, \lambda_n\} \to \{\lambda_1, \ldots, \lambda_n\}$). All in all, σ gives rise to a K-automorphism on E when dealing with field extensions in which irreducible polynomials in it split.

Theorem. Let F/K be an algebraic extension. Fix an algebraic closure $\bar{K}$ of K with $K \subset F \subset \bar{K}$. The following are equivalent:

(i) There exists a subset $S \subset K[t]$ such that F is a splitting field for S.

(ii) If $\sigma : F \to \bar{K}$ is a K-embedding, then $\sigma(F) = F$.

(iii) F/K is normal.

Proof. Assume that condition (ii) holds. To show that F is normal over K, let α be an element of F. Suppose we are given a zero β of $\mathrm{irr}(\alpha, K)$ in $\bar{K}$. We wish to show that $\beta \in F$.

First we have a K-isomorphism $\sigma_0 : K(\alpha) \to K(\beta)$. Using the isomorphism extension theorem, we can extend σ to a K-embedding $\sigma : F \to \bar{K}$. By our assumption, $\sigma(F) = F$. In particular, $\beta = \sigma(\alpha) \in F$.

$$
\begin{tikzcd}
F \arrow[r, "\sigma"] & \bar{K} \\
K(\alpha) \arrow[u] \arrow[r, "\sigma_0"] & K(\beta) \arrow[u] \\
K \arrow[u] \arrow[r] & K \arrow[u]
\end{tikzcd}
$$

$\square$

Lemma. Let F/K be an algebraic extension, and let $\sigma : F \to F$ be a K-embedding. Then $\sigma(F) = F$.

Proof. Let $\alpha_0 \in F$ be an element. Set $\text{irr}(\alpha_0, K) = (t-\alpha_0)(t-\alpha_1)\dots(t-\alpha_r)(t-\beta_1)\dots(t-\beta_s) \in \bar{K}[t]$ where $\alpha_i \in F$ and $\beta_j \in \bar{K}\backslash F$. Then σ permutes $\{\alpha, \alpha_1, \dots, \alpha_r\}$ so that $\alpha_0 = \sigma(\alpha_i)$ for some $0 \le i \le r$. $\square$

Remark. Let F/K be a finite extension. Then F/K is normal if and only if

$$[F : K]_s = |\text{Aut}(F/K)|$$

Lecture 16 (November 6th)

Remark. Last class we have learned normal extensions. This class, we learn more about normal extensions and newly finite fields.

Theorem. Let F/K be an algebraic extension and $\bar{K}$ be an algebraic closure of K with $F \subset \bar{K}$. The following are equivalent:

(i) There exists $S \subset K[t]$ such that F is a splitting field for S (that is, $F = K(S_0)$ where S_0 is the set of zeros of $f(t) \in S$).

(ii) For each K-embedding $\sigma : F \to \bar{K}$, $\sigma(F) = F$.

(iii) F/K is normal. That is, for each $\alpha \in F$, all zeros of $\text{irr}(\alpha, K)$ in $\bar{K}$ lie in F.

Example. (14.13) Consider $p(t) = t^4+t^3+t^2+t+1 \in \boldsymbol{Q}[t]$. Let $E = \boldsymbol{Q}(\xi)$, ξ is a primative 5th root of unity. Observe that σ_i can be defined to send any $\xi \mapsto \xi^i$ for $1 \le i \le 4$.

$$
\begin{tikzcd}
 & \bar{\boldsymbol{Q}} \\
E = \boldsymbol{Q}(\xi) \arrow[r, "\sigma_i"] & \boldsymbol{Q}(\xi^i) = \boldsymbol{Q}(\xi) \arrow[u] \\
\boldsymbol{Q} \arrow[u] \arrow[r] & \boldsymbol{Q} \arrow[u]
\end{tikzcd}
$$

Now consider $p(t) = t^3 - 2 \in \boldsymbol{Q}[t]$. Let $E = \boldsymbol{Q}(2^{1/3}, \xi)$ where ξ is a primative 3rd root of unity.

$$
\begin{tikzcd}
F = \boldsymbol{Q}(2^{1/3}) \arrow[r, "\sigma"] & \bar{\boldsymbol{Q}} \\
\boldsymbol{Q} \arrow[u] \arrow[r] & \boldsymbol{Q} \arrow[u]
\end{tikzcd}
$$

Here, $\sigma(2^{1/3}) = 2^{1/3}\xi \notin \boldsymbol{Q}(2^{1/3})$.

Corollary. Let F/K be a finite extension. Then F is normal over K if and only if

$$[F : K]_s = |\mathrm{Aut}(F/K)|$$

where $\mathrm{Aut}(F/K)$ is the group of all K-isomorphisms $\sigma : F \to F$, $\sigma \cdot \mu = \sigma \circ \mu$.

Remark. (Observation) Choose an algebraic closure of K with $K \subset F \subset \bar{K}$. There is an injection

$$\mathrm{Aut}(F/K) \to \{\sigma : F \to \bar{K}\}$$

where σ is a K-embedding. Schematically, we have a map

$$
\left(
\begin{tikzcd}
F \arrow[rr, "\sigma"] \arrow[dr, dash] & & F \arrow[dl, dash] \\
 & K &
\end{tikzcd}
\right)
\longmapsto
\left(
\begin{tikzcd}
F \arrow[r, "\sigma"] \arrow[dr, dash] & F \arrow[r, "\iota"] \arrow[d, dash] & \bar{K} \arrow[dl, dash] \\
 & K &
\end{tikzcd}
\right)
$$

Proposition. If F/K is a field extension of degree 2, then F/K is normal over K.

Proof. Choose an element $\alpha \in F \backslash K$. Then $F = K(\alpha)$. Let $\mathrm{irr}(\alpha, K) = t^2 + at + b \in K[t]$. Note that

$$t^2 + at + b = (t - \alpha)(t + (a + \alpha)) \in \bar{K}[t]$$

We claim that F is a splitting field for $\operatorname{irr}(\alpha, K)$. This follows from the observation that $-(a+\alpha)$ and α belong in F. An observation is that for $F/E/K$, if F/K is normal, then F/E is normal. $\square$

Chapter 15. Galois Theory

Definition. (15.1) Let F/K be a field extension. Then the Galois group of F/K is the group of K-automorphisms on F. We will denote it by $\operatorname{Gal}(F/K)$.

Lecture 17 (November 12th)

Theorem. Consider F/K, an algebraic extension. The following are equivalent:

(i) $\exists S \subset K[t]$ such that F is a simplitting field for S

(ii) If $\sigma : F \to \bar{K}$ is a K-embedding, then $\sigma(F) = F$

(iii) F/K is normal

Corollary. Let F/K be a finite extension. F/K is normal if and only if

$$[F : K]_s = |\operatorname{Aut}(F/K)|$$

Proposition. If $[F : K] = 2$, then F/K is normal.

15.1 Galois Groups and Galois Extensions

Definition. Let F/K be a field extension. The Galois group of F/K, denoted $\operatorname{Gal}(F/K)$ is the group of isomorphisms $\sigma : F \to F$ that resticts to the identity on K.

Definition. (15.2) Let F/K be an algebraic extension. We will say that F/K is Galois if it is separable and normal.

Proposition. Let F/K be a finite extension. The following are equivalent:

(i) F/K is Galois

(ii) $[F : K] = |\operatorname{Aut}(F/K)|$

(iii) F is a splitting field of a separable polynomial in $K[t]$

Proof. We prove (i) if and only if (ii). Note that $|\operatorname{Aut}(F/K)| \leq [F : K]_s \leq [F : K]$. The first equality holds if and only if F/K is normal and the second holds if and only if F/K is separable.

We now prove (i) to (iii). By the primative element theorem, we can wriet $F = K(\alpha)$ for some $\alpha \in F$. Set $f(t) = \mathrm{irr}(\alpha, K) \in K[t]$. Since F is normal over K, F is splitting field of $f(t)$. We note that $f(t)$ is separable since F/K is separable.

Finally, (iii) to (i). Suppose that $F = K(\alpha_1, \dots, \alpha_n)$ is a splitting field of a separable polynomial $f(t)$. Using the normal extension theorem, we see that F is normal over K. Note that each α_i is separable over K. This implies that $K(\alpha_1, \dots, \alpha_n) = F$ is separable. $\square$

Definition. (15.15) Let F/K be a field extension. Let G be a subgroup of $\mathrm{Gal}(F/K)$. The fixed field of G is the sufield:

$$\{\alpha \in F \mid \sigma(\alpha) = \alpha \text{ for all } \sigma \in G\} \subset F$$

denoted F^G.

Remark. (Observation) F^G is indeed a subfield of F containing K ($K \subset F^G \subset F$).

Remark. (Remark) Let $F/E/K$ be field extensions. Then $\mathrm{Gal}(F/E)$ is a subgroup of $\mathrm{Gal}(F/K)$.

Theorem. (15.31 + 15.35) (The main theorem of Galois theory) Let F/K be a finite Galois extension. Then there is an order-reversing bijection between the set of all intermediate fields for F/K and the set of all subgroups of $\mathrm{Gal}(F/K)$.

Lecture 18 (November 13th)

Remark. Last time we learned Galois theory. Today, we learn the main theorem and examples of $\mathrm{Gal}(F/K)$.

Definition. (15.2) An algebraic extension F/K is Galois if it is separable and normal.

Remark. A finite extension F/K is Galois if and only if F is splitting field of a separable polynomial in $K[t]$ if and only if

$$[F : K] = |\mathrm{Aut}(F/K)| = |\mathrm{Gal}(F/K)|$$

Remark. Let F/K is finite

$$|\mathrm{Aut}(F/K)| \leq [F : K]_s \leq [F : K]$$

the first equality holds if and only if F/K is normal and the second equality holds if and only if F/K is separable. We see that the two conditions we previously sought were exactly F being normal and separable.

Theorem. (15.31) and (15.35) (Fundamental theorem of Galois theory) Let F/K be a finite Galois extension. Then the Galois correspondence

$$\{E : \text{ an intermediate field } K \subset E \subset F\} \longleftrightarrow \{\text{subgroups of } \mathrm{Gal}(F/K)\}$$

where $E \mapsto \mathrm{Gal}(F/E)$ and $H \mapsto F^H$ is a bijection.

Proof. If E is an intermediate field $K \subset E \subset F$, then the extension E/K is Galois (or normal) if and only if $\mathrm{Gal}(F/E)$ is a normal subgroup of $\mathrm{Gal}(F/K)$. In this case, $\mathrm{Gal}(E/K)$ is isomorphic to $\mathrm{Gal}(F/K)/\mathrm{Gal}(F/E)$. $\square$

Example. (15.12) Let $f(t) = t^4 + t^3 + t^2 + t + 1 \in \boldsymbol{Q}[t]$. Let $F =$ the splitting field of $f(t)$, equal to $\boldsymbol{Q}(\xi, \xi^2, \xi^3, \xi^4)$. Here, ξ is a primative 5th root of unity. Note that

(i) $F/\boldsymbol{Q}$ is finite Galois.

(ii) Let $\sigma \in \mathrm{Gal}(F/\boldsymbol{Q})$.

$$\begin{tikzcd}
\boldsymbol{Q}(\xi, \xi^2, \xi^3, \xi^4) \arrow[rr, "\sigma"] \arrow[dr] & & \boldsymbol{Q}(\xi, \xi^2, \xi^3, \xi^4) \arrow[dl] \\
& \boldsymbol{Q} &
\end{tikzcd}$$

(ii-i) It is enough to consider $\sigma(\xi)$

(ii-ii) $\sigma(\xi)$ must be zero of $f(t)$ ($f(\xi) = 0$ implies that $f(\sigma(\xi)) = 0$) which implies that $\sigma(\xi) = \xi^i$ for $1 \leq i \leq 4$.

(ii-iii) There exists a $\sigma_i \in \mathrm{Gal}(F/K)$ such that $\sigma_i(\xi) = \xi^i$ for each $1 \leq i \leq 4$ by the isomorphism extension theorem.

(ii-iv) $\mathrm{Gal}(F/\boldsymbol{Q}) = \{\sigma_1, \sigma_2, \sigma_3, \sigma_4\}$

(ii-v) $(\boldsymbol{Z}/5\boldsymbol{Z})^\times \to \mathrm{Gal}(F/\boldsymbol{Q})$.

Lecture 19 (November 18th)

Remark. Last time, we saw examples of $\mathrm{Gal}(F/K)$. Today, we learn more about the examples and we also learn Galois theory. Note that regarding notation, the textbook uses $\mathrm{Gal}_K(F)$. However, in our class, we use $\mathrm{Gal}(F/K)$.

Example. (15.11) Let $f(t) = t^4 + t^3 + t^2 + t + 1 \in \boldsymbol{Q}[t]$, and let F be the splitting field of $f(t)$, $\boldsymbol{Q}(\xi, \xi^2, \xi^3, \xi^4) = \boldsymbol{Q}(\xi)$ (where ξ is the primative 5th root of unity). Calculate $\mathrm{Gal}(F/\boldsymbol{Q})$.

Proof. (i) For each $n \in \{1, 2, 3, 4\}$, $\sigma_{\bar{n}} : \boldsymbol{Q}(\xi) \to \boldsymbol{Q}(\xi) : \xi \mapsto \xi^n$ is a element of $\mathrm{Gal}(F/\boldsymbol{Q})$.

(ii) There is a map $\phi : (\boldsymbol{Z}/5\boldsymbol{Z})^\times \to \mathrm{Gal}(F/\boldsymbol{Q})$ that maps $\bar{n} \mapsto \sigma_{\bar{n}}$. Here, we regard $((\boldsymbol{Z}/5\boldsymbol{Z})^\times, +)$ as an albelian group.

(iii) ϕ is an isomorphism and in conclusion, $\mathrm{Gal}(F/\boldsymbol{Q})$ is a cyclic group of order 4.

Example. (15.13) Now lets turn to $f(t) = t^3 - 2$. Suppose $\sigma \in \mathrm{Gal}(F/\boldsymbol{Q})$. An important observation is that for each zero $*$ of $f(t)$, $\sigma(*)$ is a zero of $f(t)$ (note that $* \in \{\alpha, \alpha\xi, \alpha\xi^2\}$). Therefore, $\sigma(\alpha) \in \{\alpha, \alpha\xi, \alpha\xi^2\}$. Now we determine $\sigma(\alpha\xi)$. Then

$$\sigma(\alpha\xi) \in \{\alpha, \alpha\xi, \alpha\xi^2\} - \{\sigma(\alpha)\}$$

and also

$$\sigma(\alpha\xi^2) \in \{\alpha, \alpha\xi, \alpha\xi^2\} - \{\sigma(\alpha), \sigma(\alpha\xi)\}$$

The conclusion is that σ restricts to a bijection

$$\{\alpha, \alpha\xi, \alpha\xi^2\} \to \{\alpha, \alpha\xi\alpha\xi^2\}$$

Conversely, for each bijection, we can obtain a element of $\mathrm{Gal}(F/\boldsymbol{Q})$ (since $F/\boldsymbol{Q}$ is Galois). It turns out that the map $\mathrm{Gal}(F/\boldsymbol{Q}) \to S_4$, $\sigma \mapsto (\{\alpha, \alpha\xi, \alpha\xi^2\} \to \{\alpha, \alpha\xi, \alpha\xi^2\})$ is an isomorphism.

Remark. In example (15.12), we got a group homomorphism

$$\mathrm{Gal}(F/\boldsymbol{Q}) \to S_4$$

where

$$\sigma \mapsto (\{\xi, \xi^2, \xi^3, \xi^4\} \to \{\xi, \xi^2, \xi^3, \xi^4\})$$

Lemma. (15.8) Let F/K be an algebraic extension. Suppose that $F = K(\alpha_1, \ldots, \alpha_n)$, where α_i's are the zeros in F of a polynomial $f(t) \in K[t]$. Then $\mathrm{Gal}(F/K)$ is isomorphic to a subgroup of S_n.

Proof. Each σ determined by the values at α_i. In fact, there is a group homomoprhism

$$\mathrm{Gal}(F/K) \to S_{\{\alpha_1, \ldots, \alpha_n\}}$$

Moreover, this map is injective. For this, we will use group actions. $\square$

Lecture 20 (November 20th)

Remark. Last class, we have seen the two paritcular examples $\mathrm{Gal}(\boldsymbol{Q}(\xi), \boldsymbol{Q}) \cong \boldsymbol{Z}/4\boldsymbol{Z}$ and also $\mathrm{Gal}(\boldsymbol{Q}(\alpha, \alpha\xi, \alpha\xi^2)/\boldsymbol{Q}) \cong S_3$. Today, we will be learning Galois theory.

Corollary. (15.8) Let $f(t)$ be a separable polynomial in $K[t]$, and let $F = K(\alpha_1, \dots, \alpha_n)$ be the splitting field of $f(t)$. Then the Galois group $\mathrm{Gal}(F/K)$ is isomorphic to a subgroup of S_n.

Proof. Let $\sigma \in \mathrm{Gal}(F/K)$ be an element. Then for each α_i, its image under σ is a zero of $f(t)$: In fact, σ determines a bijection $\{\alpha_1, \dots, \alpha_n\} \to \{\alpha_1, \dots, \alpha_n\}$. Moreover, we have a group homomorphism

$$\phi : \mathrm{Gal}(F/K) \to S_n$$

Since $F = K(\alpha, \dots, \alpha_n)$, ϕ is injective. □

Theorem. (15.31) (Fundamental theorem of Galois theory I) Let F/K be a finite Galois extension. Then the Galois correspondence is a bijection: that is, there is a bijection between the intermediate fields $K \subset E \subset F$ and the subgroups H of $\mathrm{Gal}(F/K)$, given by $E \mapsto \mathrm{Gal}(F/E)$ and $H \mapsto F^H = \{\alpha \in F \mid \sigma(\alpha) = \alpha \text{ for all } \sigma \in H\}$.

Proof. We prove only one direction. We first show that the map $E \mapsto \mathrm{Gal}(F/E)$ is injective. For this, it suffices to show that $F^{\mathrm{Gal}(F/E)} = E$ for each subfield $K \subset E \subset F$. For this, observe that $E \subset F^{\mathrm{Gal}(F/E)}$ as for any $\alpha \in E$, $\sigma \in \mathrm{Gal}(F/E)$ satisfies $\sigma(\alpha) = \alpha$, and $\alpha \in F^{\mathrm{Gal}(F/E)}$.

For the other side, note that $F/F^{\mathrm{Gal}(F/E)}$ is a finite Galois extension and that $\mathrm{Gal}(F/F^{\mathrm{Gal}(F/E)})$ is a subgroup of $\mathrm{Gal}(F/E)$. Since $\mathrm{Gal}(F/E)$ is a subgroup of $\mathrm{Gal}(F/F^{\mathrm{Gal}(F/E)})$, we see that $\mathrm{Gal}(F/E) = \mathrm{Gal}(F/F^{\mathrm{Gal}(F/E)})$. To conclude, we observe that

$$[F : F^{\mathrm{Gal}(F/E)}] = |\mathrm{Gal}(F/F^{\mathrm{Gal}(F/E)})| = |\mathrm{Gal}(F/E)| = [F : E]$$

Thus $[F^{\mathrm{Gal}(F/E)} : E] = 1$ and $F^{\mathrm{Gal}(F/E)} = E$.

To prove the surjectivity (of $E \mapsto \mathrm{Gal}(F/E)$), let H be a subgroup of $\mathrm{Gal}(F/K)$. We will show that $H = \mathrm{Gal}(F/F^H)$. For this, let $\alpha \in F$ be an element. Take a maximal subset $\{\sigma_1, \dots, \sigma_r\} \subset H$ such that $\{\sigma_1(\alpha), \dots, \sigma_r(\alpha)\}$ are mutually distinct. We may assume that $\sigma_1 = 1_F$. We may assume that $\sigma_1 = 1_K$. Let

$$f(t) = \prod_{i=1}^{r} (t - \sigma_i(\alpha))$$

□

Lecture 21 (November 25th)

Remark. Last class, we learned Galois theory and its proof. Today, we will continue the proof and look at some examples.

Theorem. (15.31) (Fundamental theorem of Galois theory I) Let F/K be a finite Galois extension.

$$\begin{aligned}\{K \subset E \subset F\} &\leftrightarrow \{1 \subset H \subset \mathrm{Gal}(F/K)\}\\ E &\mapsto \mathrm{Gal}(F/E)\\ H &\mapsto H\end{aligned}$$

Proof. The injectivity of $E \mapsto \mathrm{Gal}(F/E)$ follows for any intermediate field E. To prove that the map

$$E \mapsto \mathrm{Gal}(F/E)$$

is surjective, let H be a subgroup of $\mathrm{Gal}(F/K)$, we claim that $\mathrm{Gal}(F/F^H) = H$. Let $\alpha \in F$ be an element. Take a maximal subset $\{\sigma_1, \dots, \sigma_r\} \subset H$ such that $\sigma_1(\alpha), \dots \sigma_r(\alpha)$ are mutually distinct. We may suppose that $\sigma_1 = 1_F$. Let

$$f(t) = \prod_{i=1}^{r}(t - \sigma_i(\alpha)) \in F[t] \supset F^H[t]$$

Then for each $\tau \in H$, the maximality of the subset implies that

$$\{(\tau \circ \sigma_1)(\alpha), \dots, (\tau \circ \sigma_r)(\alpha)\} \subset \{\sigma_1(\alpha), \dots, \sigma_r(\alpha)\}$$

It then follows that $f(t) \in F^H[t]$. We now complete the proof. Using the fact that $H \lhd \mathrm{Gal}(F/F^H)$, $|H| \leq |\mathrm{Gal}(F/F^H)| = [F : F^H]$. Consequently, it will suffice to show that $[F : F^H] \leq |H|$. By the primative element theorem, $F = F^H(\alpha)$ for some $\alpha \in F$. It follows from the previous argument that

$$[F : F^H] = [F^H(\alpha) : F^H] = \deg(\mathrm{irr}(\alpha, F^H)) \leq f(t) \leq |H|$$

$\square$

Remark. The proof shows the following: for each $\alpha \in F$, let $\sigma_1(\alpha), \dots, \sigma_r(\alpha)$ be the Galois conjugates of α (that is, it is the set $\{\sigma_i(\alpha) \mid \sigma_i \in \mathrm{Gal}(F/K)\}$). Then

$$f(t) = \prod_{i=1}^{r}(t - \sigma_i(\alpha)) \in F^{\mathrm{Gal}(F/K)}[t] = K[t]$$

Is $f(t) = \mathrm{irr}(\alpha, K)$? According to proposition (15.24), $f(t)$ is indeed the minimal polynomial of α over K.

Lecture 22 (November 27th)

Remark. Last class, we have learned the fundamental theorem of Galois theory I. Today, we will look at some examples and the concept of group action.

Theorem. (15.31) For a finite Galois extension F/K, there exists a correspondence between the Galois group (the group of automorphisms) and intermediate fields. Explicitly, the map

$$\{K \subset E \subset F\} \leftrightarrow \{H \triangleleft \mathrm{Gal}(F/K)\}$$

is given by

$$E \mapsto \mathrm{Gal}(F/E)$$

and

$$H \mapsto F^H$$

Definition. (15.22) Let F/K be a finite Galois extension. Let $\alpha \in F$ be an element. The disinct $\sigma(\alpha)$ for $\sigma \in \mathrm{Gal}(F/K)$ are the Galois conjugates of α.

Proposition. (15.24) Let F/K be a finite Galois extension. Let $\alpha \in F$ be an element. Let $\alpha_1, \ldots, \alpha_r$ be the Galois conjugates of α. Then

$$f(t) = \prod_{i=1}^{r}(t - \alpha_i)$$

is the minimal polynomial of α over K.

Proof. It suffices to show that $\deg f(t) = \deg \mathrm{irr}(\alpha, K)$. Note that

$$\deg(\mathrm{irr}(\alpha, K)) = [K(\alpha) : K] = \frac{[F : K]}{[F : K(\alpha)]} = \frac{|\mathrm{Gal}(F/K)|}{|\mathrm{Gal}(F : K(\alpha))|} = [\mathrm{Gal}(F/K) : \mathrm{Gal}(F/K(\alpha))]$$

Consider the action of $\mathrm{Gal}(F/K)$ on the set of the Galois conjugates of α. Applying lemma (12.9), we see that

$$\deg f(t) = r = [\mathrm{Gal}(F/K) : \mathrm{Gal}(F/K(\alpha))]$$

$\square$

Definition. (11.9) (Group actions) Let G be a group and X be a set. An action of G on X is a function $\phi : G \times X \to X$ such that:

(i) $\phi(1, x) = x$ for all $x \in X$.

(ii) $\phi(gh, x) = \phi(g, \phi(h, x))$ for all $g, h \in G$ and all $x \in X$.

Proposition. (11.20) Giving an action of G on X is equivalent to giving a group homomorphism

$$G \to S_x$$

Proof. Ours! $\square$

Example. (i) G acts on itself by left multiplication $G \times \underline{G} \to G : (g,h) \mapsto gh$

(ii) G acts on itself by conjugation $G \times \underline{G} \to \underline{G} : (g,h) \mapsto ghg^{-1}$

Definition. (11.22) Let X be a G-set. The orbit of an element $x \in X$ is the subset $\{g \cdot x \mid g \in G\} \subset X$.

Definition. (12.8) The stabiliser of $x \in X$ is the subgroup

$$G_x = \{g \in G \mid g \cdot x = x\}$$

Proposition. Let S_n act on $\{1, \ldots, n\}$. Let $H \triangleleft S_n$. Then is $H = (S_n)_i$ for some i?

Lemma. (12.9) There is a bijection between the orbit of X and the set G/G_X of left cosets of G_X.

Proof. Define a function $\phi : G_X \to G/G_X$. Check that it is well defined and that it is bijective. $\square$

Proposition. Let K be a field, and $f(t) \in K[t]$ be a separable polynomial. The following are equivalent:

(i) $f(t) \in K[t]$ is irreducible

(ii) $\mathrm{Gal}(K(\alpha_1, \ldots, \alpha_n)/K)$ acts transistively on $\{\alpha_1, \ldots, \alpha_n\}$ where $\alpha_1, \ldots, \alpha_n$ are the zeros of $f(t)$ in $\bar{K}$. That is, for any given pair of elements (α_i, α_j), there exists an element $\sigma \in \mathrm{Gal}(K(\alpha_1, \ldots, \alpha_n)/K)$ such that $\sigma(\alpha_i) = \sigma(\alpha_j)$.

Remark. Can we prove the irreducibility of $f(t)$ on page 44 using the above proposition?

Example. (15.32) Let F be the splitting field of $t^3 - 2 \in \boldsymbol{Q}[t]$ which is equal to $\boldsymbol{Q}(\alpha_1, \alpha_2, \alpha_3) = \boldsymbol{Q}(2^{1/3}, \alpha_1\xi, \alpha_2\xi)$.

$$\mathrm{Gal}(F/\boldsymbol{Q}) \cong S_3$$

Due to the above, we can notice that

$$\begin{tikzcd}
 & 1 \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] \arrow[drr, dash] & & \\
\langle (1,2,3) \rangle \arrow[dr, dash] & \langle (1,2) \rangle \arrow[d, dash] & \langle (2,3) \rangle \arrow[dl, dash] & \langle (3,1) \rangle \arrow[dll, dash] \\
 & S_3 & &
\end{tikzcd}$$

has a bijection with

$$\begin{tikzcd}
 & F \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] \arrow[drr, dash] & & \\
F^{\langle (1,2,3) \rangle} \arrow[dr, dash] & F^{\langle (1,2) \rangle} \arrow[d, dash] & F^{\langle (2,3) \rangle} \arrow[dl, dash] & F^{\langle (3,1) \rangle} \arrow[dll, dash] \\
 & \boldsymbol{Q} & &
\end{tikzcd}$$

Letting us conclude that

$$[F : F^H] = |H|$$

Lecture 23 (December 2nd)

Remark. Last time, we learned about examples of the Galois correspondence. This class, we learn about the Galois group of polynomials. Make sure to read examples (15.32) and (15.33).

Lemma. (15.37) Let $f(t) \in K[t]$ be separable. Then, $f(t) \in K[t]$ is irreducible if and only if the Galois group $\mathrm{Gal}(K(\alpha, \ldots, \alpha_n)/K)$ acts transistively on $\{\alpha_1, \ldots, \alpha_n\}$.

Lemma. Let $f(t) \in K[t]$ and separable. Then,

(i) There exists an injection $\mathrm{Gal}(K(\alpha, \ldots, \alpha_n)/K) \hookrightarrow S_n$.

(ii) If $f(t)$ is irreducible in $K[t]$, then n divides $|\mathrm{Gal}(K(\alpha, \ldots, \alpha_n)/K)|$.

Remark. (Notation) Let $f(t) \in K[t]$ be a polynomial over a field K. We let G_f denote the Galois group of the extension $K(\alpha, \dots, \alpha_n)/K$, where $\alpha_1, \dots, \alpha_n$ are the zeros of $f(t)$ (in $\bar{K}$).

Proof. For (ii), notice that $\deg(\operatorname{irr}(\alpha_1, K)) = \deg(f(t))$ and that:

$$n = [K(\alpha_1) : K] \mid [K(\alpha, \dots, \alpha_n) : K] = |G_f|$$

$\square$

Remark. (Notation)

(i) K is a field

(ii) $x_1, \dots, x_n$ are mutually distinct indeterminants

(iii) $g(t) = \prod_{i=1}^{n}(t - x_i) = t^n - s_1 t^{n-1} + s_2 t^{n-2} + \ldots + (-1)^n s_n \in E[t]$. In particular,

$$s_1 = \sum_{1 \le i \le n} x_i \qquad s_2 = \sum_{1 \le i \le j \le n} x_i x_j \qquad s_3 = \sum_{1 \le i \le j \le k \le n} x_i x_j x_k \qquad s_n = \prod_{i=1}^{n} x_i$$

(iv) $E = K(s_1, \dots, s_n)$

(v) $F = K(x_1, \dots, x_n)$

(vi) $g(t) \in E[t]$

In this setting, F is the splitting field of $g(t) \in E[t]$ and the extension F/E is finite Galois.

Proposition. There is an isomorphism $G_g \to S_n$.

Proof. Due to the above lemma, there exists an injection $G_g \to S_n$. To prove surjectivity, we will show that there is a injection $S_n \to G_g$. For this, we note that each element of S_n induces a K-automorphism on $K[x_1, \dots, x_n]$, hence a K-automorphisms on $\operatorname{Frac}(K[x_1, \dots, x_n]) = K(x_1, \dots, x_n) = F$. It gives an injective group homorphism $S_n \to \operatorname{Gal}(F/K) \supset \operatorname{Gal}(F/E) = G_g$. Now the punchline – each $\sigma \in S_n$ fixes all s_i's due to the fact that $g(t) = g_\sigma(t)$. Consequently, we get an injection

$$S_n \to \operatorname{Gal}(F/E) = G_g$$

$\square$

Remark. $g(t) = \prod(t - x_i) \in E[t]$ is irreducible by the lemma.

Lecture 24 (December 4th)

Remark. Last time, we considered the Galois group of a general polynomial of degree n. Today, we will consider G_f for a cubic f.

Recall. We let K a field, $x_1, \ldots, x_n$ indeterminates, and

$$g(t) = \prod_{i=1}^{n}(t - x_i) = t^n - s_1 t^{n-1} + s_2 t^{n-2} + \ldots + (-1)^n s_n \in K(s_1, \ldots, s_n)$$

Define

$$E = K(s_1, \ldots, s_n) \qquad \& \qquad f(t) \in E[t]$$

and $G_g = \mathrm{Gal}(F/E)$

Proposition. (15.37) G_g is isomorphic to S_n.

Remark. Let $G_g \in E[t]$ be irreducible by the lemma.

Remark. Let $f(t) \in K[t]$ be a polynomial of degree n and let $\alpha_1, \ldots, \alpha_n$ be the zeros of $f(t)$ in $\bar{K}$.

$$\delta = \prod_{1 \leq i \leq j \leq n} (a_j - a_i) \qquad \& \qquad D = \delta^2$$

is called the discriminant of f in $\bar{K}$.

Example. (Galois group of polynomials of degree 3) We warm up by investigating $f(t) = t^2 + bt + c \in K[t]$.

(i) If f has a repeated root, $G_f = 1$

(ii) If $f(t) = (t - \alpha)(t - \beta)$, $\alpha \neq \beta$, let F be the splitting field of f. $F = K(\alpha, \beta) = K(\alpha) = K(\delta)$. In this case, $\delta_f = \beta - \alpha$ and $D = \delta_f^2 = (\beta - \alpha)^2 = b^2 - 4c$.

$$G_f = \mathrm{Gal}(F/K) = \mathrm{Gal}(K(\delta)/K) = S_2$$

Remark. Let $g(t) = t^3 - s_1 t^2 + s_2 t - s_3$ and $D = -4s_2^3 - 27s_3^2 - 4s_1^3 s_3 + s_1^2 s_2^2 + 18 s_1 s_2 s_3$ (hint, vandermonde determinant).

Recall. Let A_n be the alternating group, the group of even permutations. Notice that $A_n \triangleleft S_n$. Also,

$$A_n = \ker(S_n \to \{\pm 1\})$$

Lemma. (Transposition) If σ is a transposition, then $\sigma(\delta_g) = -\delta_g$. Here, $g(t)$ is the general polynomial of degree n.

Theorem. Let $n \geq 2$ be an integer. Then,

(i) $D_g \in E = K(s_1, \ldots, s_n)$.

(ii) If $\mathrm{char}(K) \neq 2$, $F^{A_n} = E(\delta_g)$.

We have the towers $F = K(x_1, \ldots, x_n) - F^{A_n} = E(\delta_g) - E = K(s_1, \ldots, s_n)$ and $1 - A_n - \mathrm{Gal}(F/E) = S_n$.

Proof. (I) Since S_n is generated by transpositions, if $\sigma \in S_n$, then

$$\sigma(D_g) = \sigma(\delta_g)^2 = (\pm\delta_g)^2 = D_g$$

(II) First, observe that $\delta_g \in F^{A_n}$ by the transposition lemma. That is, $F^{A_n} \supset E(\delta_g)$. Note that $\delta_g \notin E = F^{S_n}$ and that $\delta_g^2 = D \in E$. Therefore, $[E(\delta_g) : E] = 2$. To complete the proof, we will show that $[F^{A_n} : E] = 2$. Indeed,

$$[F^{A_n} : E] = \frac{[F : E]}{[F : F^{A_n}]} = \frac{|\mathrm{Gal}(F/E)|}{|\mathrm{Gal}(F/F^{A_n})|} = \frac{|S_n|}{|A_n|} = 2$$

where we considered the diagram

$$F - F^{A_n} - E(\delta_g) - E$$

$\square$

Corollary. Let K be a field with $\mathrm{char}(K) \neq 2$. Let $f(t)$ be an irreducible separable polynomial of degree $n \geq 2$. Let F be the splitting field of $f(t)$. Then,

(i) $F^{G_f \cap A_n} = K(\delta_f)$.

(ii) $G_f \triangleleft A_n$ if and only if $\delta_f \in K$.

Example. (15.41) Let K be a field with $\mathrm{char}(K) \neq 2$. Let $f(t)$ be an irreducible separable polynomial of degree 3. Then,

(i) $G_f \cong S_3$ if and only if $\delta_f \notin K$.

(ii) $G_f \cong A_3$ if and only if $\delta_f \in K$.

Remark. By the S_n lemma, $3 \mid |G_f|$ (since $f(t) \in K[t]$ is irreducible).

$$A_3 \triangleleft G_f \triangleleft S_3$$

Remark.

$$D_f = \delta_f^2 = t^3 - s_1 t^2 + s_2 t - s_3$$

Lecture 25 (Decemeber 9th)

Remark. Last time, we looked at G_f for a cubic f. Today, we will be solving polynomial equations.

Example. (15.41) For a field with characteristic not equal to 2 and $f(t) \in K[t]$, an irreducible separable polynomial of degree three, we noted that

$$G_f \cong S_3 \iff \delta_f \in K$$

and

$$G_f \cong A_3 \iff \delta_f \notin K$$

For $g(t) = t^3 - s_1 t^2 + s_2 t - s_3$,

$$D_g = \delta^2 = -4s_2^3 - 27s_3^2 - 4s_1^3 s_3 + s_1^2 s_2^2 + 18 s_1 s_2 s_3$$

and after applying the change of variables, one can remove all terms insolving s_1. In this case,

$$D_g = -4s_2^3 - 27s_3^2$$

Chapter 15.5 Solving Polynomial equations by Radicals

Definition. (15.5) A field extension F/K is a radical if there exists $u_1, \dots, u_r \in F$ and positive integers $m_1, \dots, m_r$ such that

$$K \subset K(u_1) \subset K(u_1, u_2) \subset \dots \subset K(u_1, \dots, u_r) = F$$

and $u_i^{m_i} \in K(u_1, u_2, \dots, u_{i-1})$ for all i. A polynomial $f(t) \in K[t]$ is solvable by radicals if its splitting field is contained in a radical extension.

Example. Let $\operatorname{char} K = 0$. Suppose $f(t) = t^2 + bt + c \in \boldsymbol{Q}[t]$. Then,

$$t = \frac{-b \pm \sqrt{b^2 - 4c}}{2}$$

and $D = b^2 - 4c$. The splitting field of $f(t)$ is $\boldsymbol{Q}(\sqrt{D})/\boldsymbol{Q}$. Thus, the function is solvable by radicals.

Theorem. (15.47) Let K be a field of $\operatorname{char} 0$ and let $f(t) \in K[t]$ be a nonconstant polynomial. Then, $f(t)$ is solvable by radicals if and only if the Galois group G_f is solvable.

Definition. (12.53) Let G be a group. Then G is solvable if there exists a tower

$$1 = G_0 \trianglelefteq G_1 \trianglelefteq G_2 \trianglelefteq \dots \trianglelefteq G_k = G$$

(that is, G_i is normal in G_{i+1}) such that G_{i+1}/G_i is abelian.

Example. (12.54 - 12.59)

(i) Albelian groups G are solvable ($0 \trianglelefteq G$).

(ii) D_{2n} is solvable ($1 \trianglelefteq \ldots \trianglelefteq D_{2n}$).

(iii) $S_3 \cong D_6$ and is solvable.

(iv) S_4 is also solvable (use the Klein 4-group)

Proposition. (12.60) Subgroups and homomorphic images of solvable groups are solvable.

Definition. Let F/K be a finite Galois extension. Then F/K is solvable if $\mathrm{Gal}(F/K)$ is solvable. Also, F/K is abelian if $\mathrm{Gal}(F/K)$ is abelian.

Corollary. (Solvable subextension) Let $K \subset E \subset F$ be solvable extensions. If F/K is solvable and E/K is Galois, then E/K is solvable.

Lemma. (Solvable extensions) Let F/K be a finite Galois extension. The following are equivalent:

(i) F/K is solvable (that is, $\mathrm{Gal}(F/K)$ is solvable)

(ii) There exists a tower $K = F_0 \subset F_1 \subset \ldots \subset F_s = F$ such that each extension F_{i+1}/F_i is abelian.

Proof. From (i) to (ii), we know that F/K is solvable and that there exists a tower,

$$1 = G_0 \trianglelefteq G_1 \trianglelefteq \ldots \trianglelefteq G_s = \mathrm{Gal}(F/K)$$

where G_{i+1}/G_i are all abelian. We acknowledge that:

$$F = F^1 = F^{G_0} \supset \ldots \supset F^{G_5} = F^{\mathrm{Gal}(F/K)} = K$$

□

Lecture 26 (December 11th)

Remark. Last time, we defined solvable by radicals. Today, it's showtime!

Definition. (15.42)

(i) A field extension F/K is radical if there exists $K \subset K(u_1) \subset K(u_1, u_2) \subset \ldots \subset K(u_1, \ldots, u_r) = F$ with $u_i^{m_i} \in K(u_1, \ldots, u_{i-1})$.

(ii) $f(t) \in K[t]$ is solvable by radicals if its splitting field is contained in some radical extension.

Definition. (12.53) A group is solvable if there exists

$$1 = G_0 \trianglelefteq G_1 \trianglelefteq \dots G_r = G$$

with G_{i+1}/G_i is abelian.

Corollary. (Solvable extensions) If $F/E/K$ is solvable and E/K is Galois, E/K is solvable.

Definition. Let F/K be finite Galois. F/K is solvable (abelian) if $\mathrm{Gal}(F/K)$ is solvable (abelian).

Lemma. (Solvable extensions) Let F/K be finite Galois. The following are equivalent:

(i) F/K is solvable.

(ii) Exists a tower
$$K = F_0 \subset F_1 \subset \dots \subset F_s = F$$
with F_{i+1}/F_i is abelian.

Lemma. (15.45) Every separable radical extension is contained in a Galois radical extension.

Proof. Let F/K be separable and radical.

$$K \subset K(u_1) \subset K(u_1, u_2) \subset \dots \subset K(u_1, \dots, u_r) = F$$

and $u_i^{m_i} \in K(u_1, \dots, u_{i-1})$. As the extension is finite and separable, by the primative element theorem, $F = K(\alpha)$. Set $p(t) = \mathrm{irr}(\alpha, K)$, and it has the zeros $\alpha = \alpha_1, \dots, \alpha_s \in \bar{K}$. We claim that $L = K(\alpha_1, \dots, \alpha_s)$, the splitting field of $f(t) \in K[t]$, is a radical extension of K. We verify that:

(i) Notice that
$$K \subset K(\alpha_1) \subset K(\alpha_1, \alpha_2) \subset \dots \subset K(\alpha_1, \dots, \alpha_s) = L$$

(ii) In each step,
$$K(\alpha_1, \dots, \alpha_{i-1}) \subset K(\alpha_1, \dots, \alpha_i)$$
is radical.

(iii) We use the fact that there exists a K-isomorphism $K(\alpha) \cong K(\alpha_i)$ is radical.

□

Lemma. (Primative) Let K be a field of characteristic zero. Let ξ be a primative n-th root of unity. Then,

$$\operatorname{Gal}(K(\xi)/K) \cong (\boldsymbol{Z}/n\boldsymbol{Z})^{\times}$$

which is exercise 15.4.

Lemma. (15.28) (Cyclic) Let K be a field containing all n-th roots of unity. Let $f(t) = t^n - a \in K[t]$ be a polynomial. Suppose $\alpha^n = a$ for some $\alpha \in \bar{K}$. Then $K(\alpha)$ is the splitting field of $f(t)$ and $G_f = \operatorname{Gal}(K(\alpha)/K)$ is cyclic.

Proof. Note that the zeros are $\alpha, \alpha\xi, \ldots, \alpha\xi^{n-1}$. Then,

$$K(\alpha, \alpha\xi, \ldots, \alpha\xi^{n-1}) = K(\alpha)$$

Notice how there exists a injection from $\operatorname{Gal}(K(\alpha)/K)$ into $\{\beta \in K \mid \beta^n = 1\}$ which is cyclic. A subgroup of a cyclic group is cyclic and the proof is complete. □

Theorem. (15.46) Let K be a field of characteristic zero. Let $f(t) \in K[t]$ be a polynomial of degree ≥ 1. Then $f(t)$ is solvable by radicals if and only if G_f is solvable.

Proof. Let F be the splitting field of $f(t)$, and let L be a radical extension of K containing F. By lemma (15.45), we may assume that L/K is Galois. Choose a tower

$$K \subset K(\alpha_1) \subset K(\alpha_1, \alpha_2) \subset \ldots \subset K(\alpha_1, \ldots, \alpha_s) = L$$

such that $\alpha_i^{n_i} \in K(\alpha_1, \ldots, \alpha_{i-1})$. Put $n = \operatorname{lcm}(n_1, \ldots, n_s)$. Let ξ be a n-th root of unity. We now consider $L(\xi)/K$. Note that $K(\xi)$ contains all primative n_i-th roots of unity. Choose $g(t) \in K[t]$ such that L/K is the splitting field for $g(t)$. Then $L(\xi)$ is the splitting field of $(t^n - 1)g(t)$. By corollary (solvable extensions), it suffices to show that $L(\xi)/K$ is solvable. Consider the tower:

$$K \subset K(\xi) \subset K(\xi, \alpha_1) \subset K(\xi, \alpha_1, \alpha_2) \subset \ldots \subset K(\xi, \alpha_1, \ldots, \alpha_s) = L(\xi)$$

Using lemma (solvable extensions), it will suffice to show that each step is an abelian extension. For this, we see that $K(\xi)/K$ is abelian by lemma (primative). Other steps are abelian by lemma (cyclic). □

Corollary. (15.48) There are no formulas in radicals for the solutions of a general polynomial equation of degree $n \geq 5$.

Proof. For $n \geq 5$, S_n is not solvable. □

Remark. You deserve a Bachelor's in math.